SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

89/100

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Duration: 60 Minutes **Total Marks: 20**

Code of Conduct

I R.Ram Nikesh (192511302) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________

Criteria	Concept Identification (1.25)	Relationship Mapping (1.25)	Hierarchy and Organization (1)	Technical Relevance (1)	Visual Clarity (0.5)	TOTAL (5)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						

Signature of the Faculty **Total Marks Awarded:**

Benefits vs challenges : Sh

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Architecture: Include front-end, back-end, client devices, data centers, and distributed servers.
2. Concept Map 2: Characteristics of Cloud Computing: Connect on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.
3. Concept Map 3: Deployment Models: Map public cloud, private cloud, hybrid cloud, and community cloud with usage scenarios.
4. Concept Map 4: Cloud Benefits vs Challenges: Show advantages (cost efficiency, scalability, flexibility) and limitations (security, downtime, vendor lock-in).

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

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Course : Cloud Computing and Big data analytics

Code : CSA1521

date : 16.7.2026

Assessement Tool : 2

(concept mapping)

1. Concept map 1 : cloud computing Architecture : Include front-end Back-end, client devices, data centers, and distributed servers.

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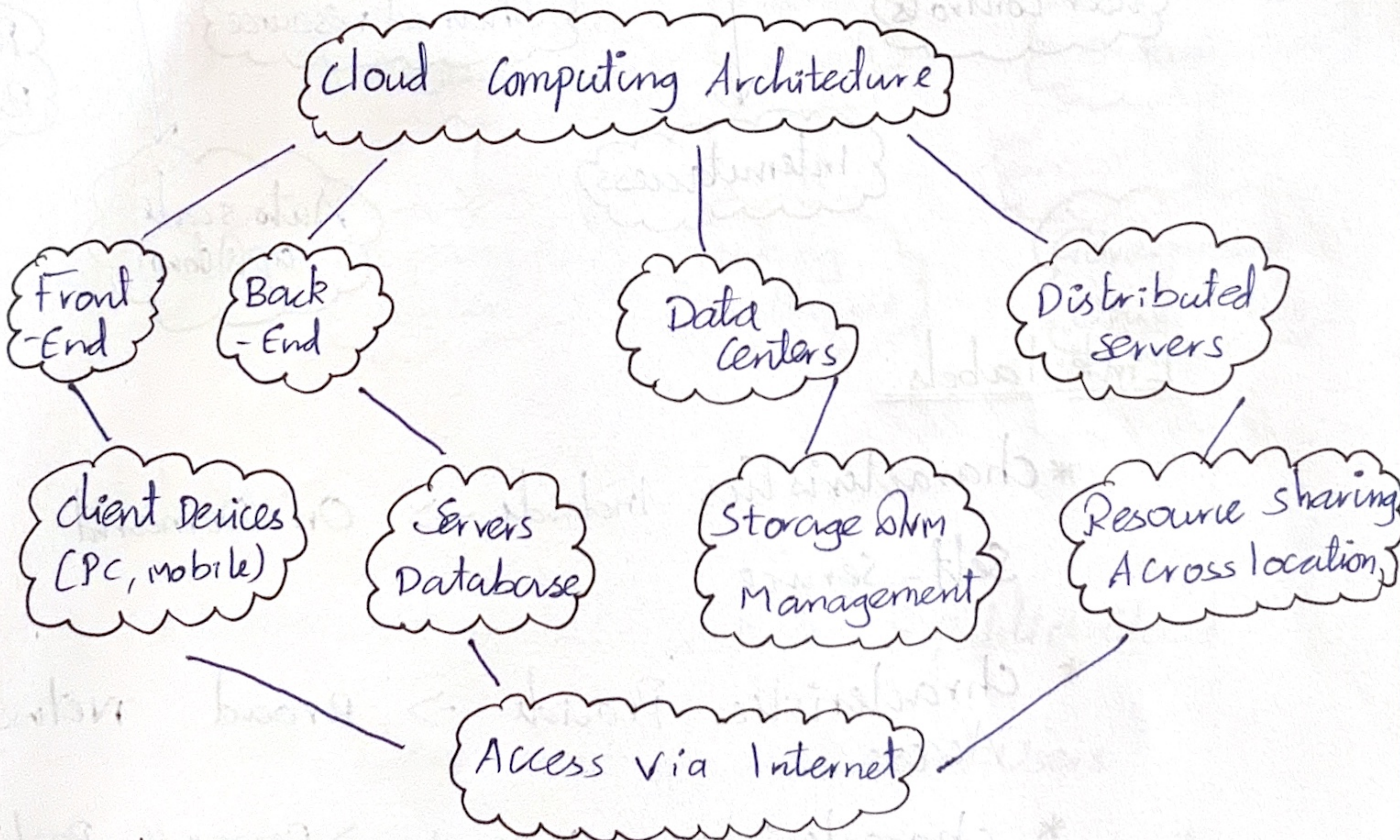

Link lables :

* Architecture Consists of -> Front-End, Back-End
* Front-End uses -> client devices
* Back-End Contains -> servers, databases
* Back-End runs in -> Data Centers
* Data Centers connect through -> distributed Servers.

2. Concept Map 2 : characteristics of cloud computing : Connect demand Self - Service , broad network access , resource pool rapid elasticity and measured service .
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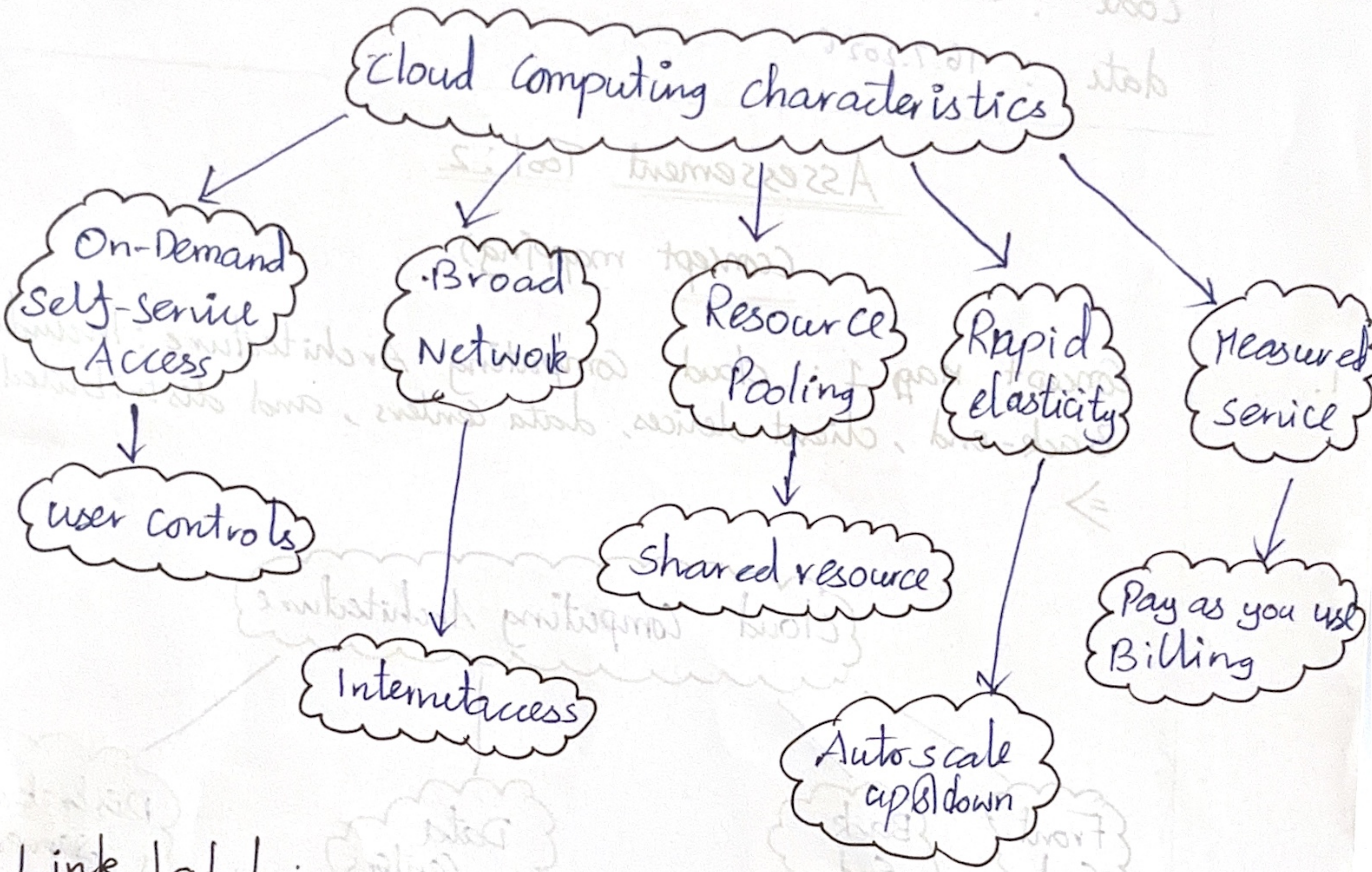

Link labels :

* Characteristics Include → On - Demand Self - Service
* Chracteristic Provide → Broad Network Access
* characteristics Support → Resource Pooling
* characteristics enable → Rapid Elasticity
* characteristic mesure → Service usage

R.Ram Nikesh

Concept Map3: Deployment Models: Map Public cloud, Private clou
hybird cloud and community cloud with usage scenar
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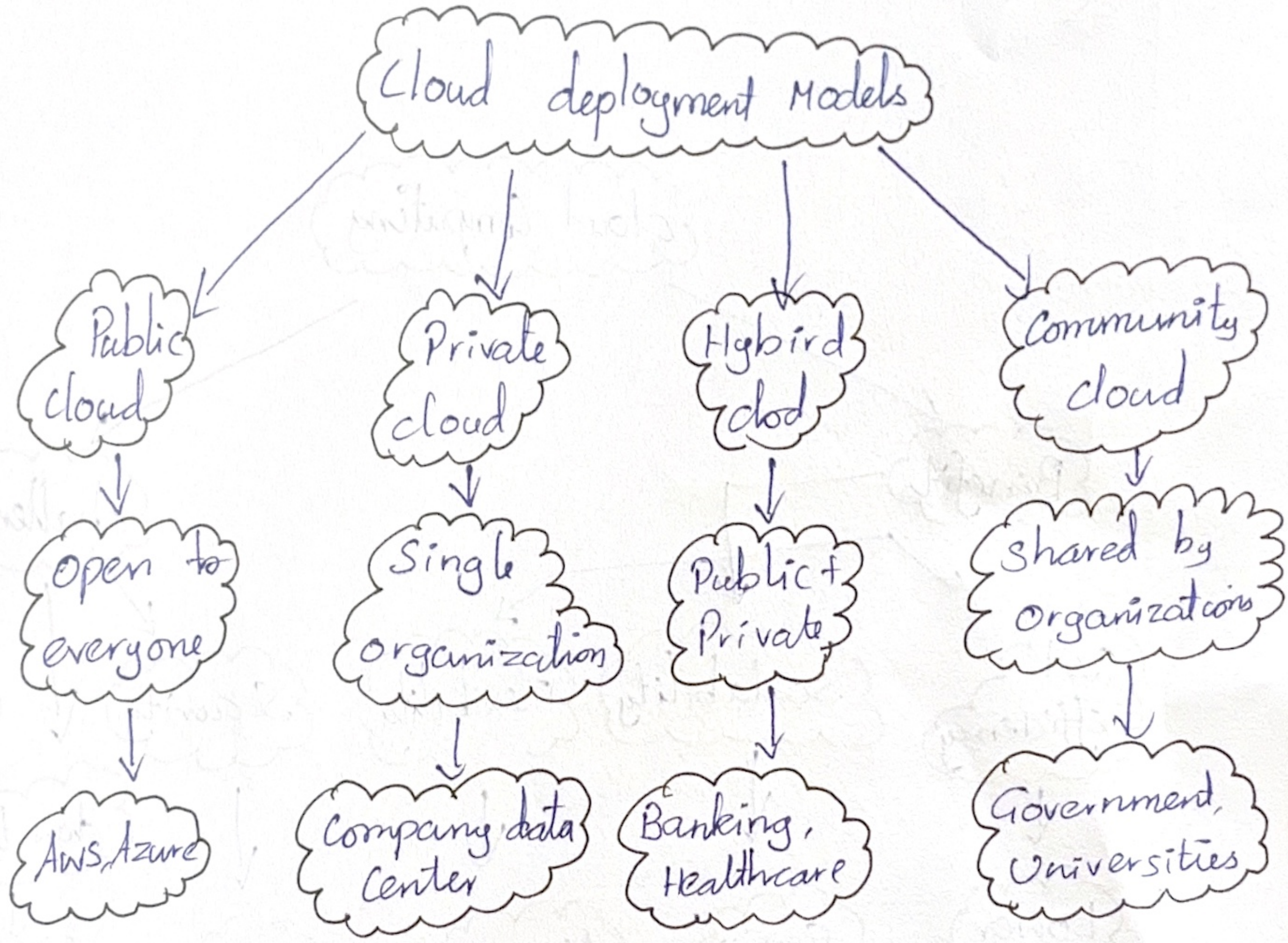

Link labels:

* Deployment models include -> Public Cloud
* Public cloud used by -> General Users
* Private cloud owned by -> one organization
* Hybird cloud combines -> Public + Private
* Community cloud shared among -> Similar organizations

4. Concept Map 4: Cloud benefits vs Challenges: Show advantages and limitations (cost efficiency, scalability, flexibility) & (Security, downtime, vendor lock-in)

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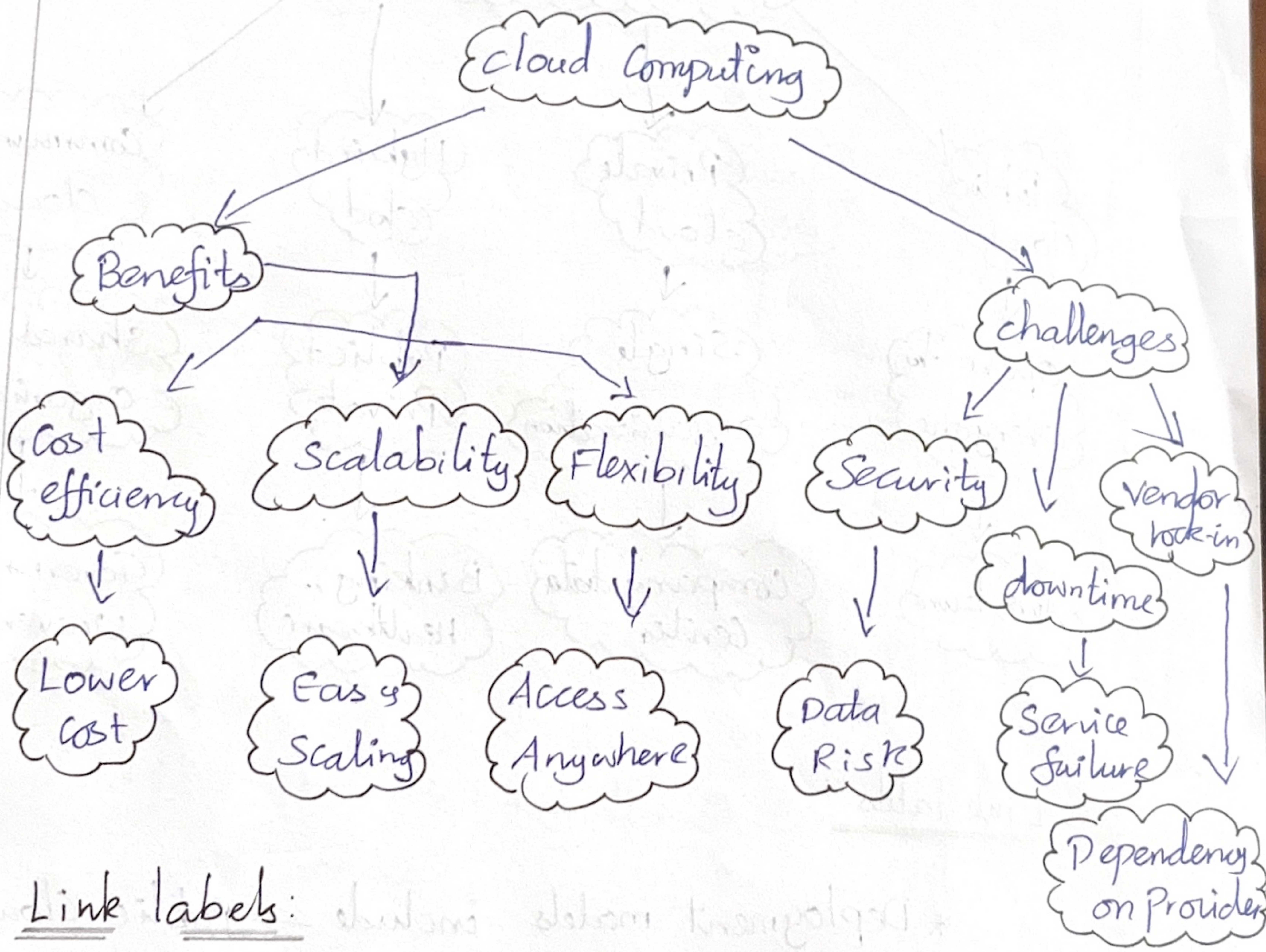

Link labels:

* Cloud computing Provides -> Benefits
* Benefits include -> cost Efficiency, Scalability, Flexibility
* Cloud computing faces -> challenges
* challenges include -> Security, Downtime, vendor lock-in